FINAL PROJECT

Ministry of Communications and Information Technology

NTI
National Telecommunication Institute

Roaya رؤية

Smart City Transformation

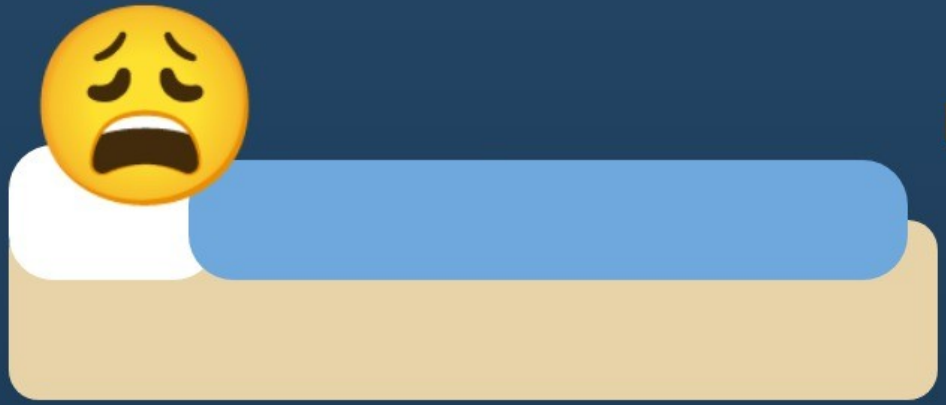
A city that saves your time, your money, and your planet.



Meet Ahmed 👋

Monday • 7:00 AM





RING RING!



OUR TEAM

Six managers, one vision



Mai Nasser Abubakr

Project Manager

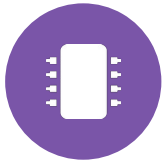
Charter & objectives



Mariam Mahmoud Dabeesh

Planning Manager

Schedule & milestones



Rahma Khaled Ahmed

Technical Manager

Technology & resources



Moaz Hamed Ibrahim

Finance Manager

Budget & funding



Omnia Sayed Abdelhakim

Risk & Quality Manager

Risks & quality



Karim Mohamed Gad

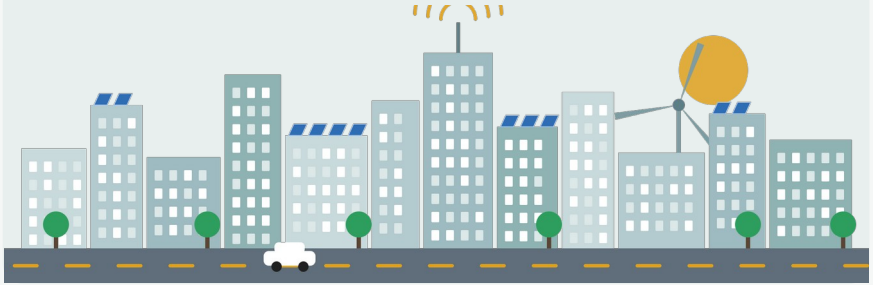
Stakeholder Manager

Stakeholders & communication

Project Manager
Mai Nasser Abubakr

Why Roaya?

Make city life...



Easier
Less time in traffic

Safer
Faster emergency help

Greener
Clean solar energy

24 months

550M EGP

7 phases

6 deliverables

OBJECTIVES

What we will achieve

Project Manager
Mai Nasser Abubakr

−30%
traffic jams

−20%
energy & water

15 min
to anywhere

50%
solar power

8 min
emergency response

20,000
new jobs

Smart
security

80%
services online

SCOPE AND DELIVERABLES

Six things we will deliver

Project Manager
Mai Nasser Abubakr



D1

Current state assessment

Mar 2027



D2

Roadmap & business case

Jun 2027



D3

Integration architecture

Sep 2027



D6

Security & data governance

Sep 2027



D4

Pilot & KPI report

Sep 2028



D5

Scale-up & change plan

Oct 2028

Planning Manager
Mariam Mahmoud Dabeesh

Smart cities are already real



EGYPT

New Administrative Capital

- One central control center
- 6,000+ smart cameras planned (2019)



SAUDI ARABIA

NEOM

- A 170 km city with no cars, planned
- Now scaled back: big plans need phases



SINGAPORE

Smart Nation

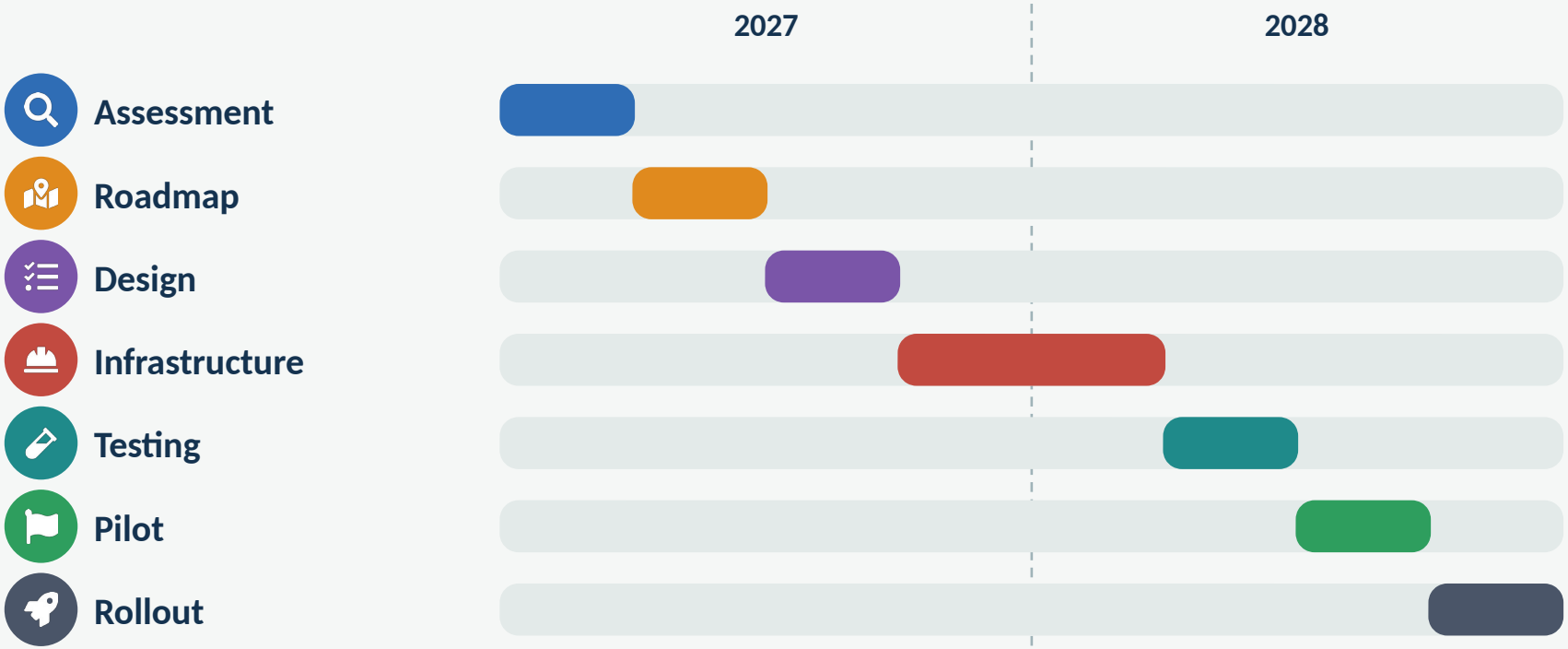
- National plan since 2014
- Government services on your phone

Photos via Wikimedia Commons: Abdelrhman 1990 (CC BY-SA 4.0) · NASA, ISS crew (public domain) · Basile Morin (CC BY-SA 4.0)

PROJECT SCHEDULE

24 months, 7 phases

Planning Manager
Mariam Mahmoud Dabeesh



MILESTONES

Planning Manager
Mariam Mahmoud Dabeesh

Eight checkpoints to the finish

Mar 2027
Assessment

Jun 2027
Roadmap

Sep 2027
Architecture & security

Mar 2028
Infrastructure ready

Jun 2028
Systems tested

Sep 2028
Pilot report

Oct 2028
Scale-up plan

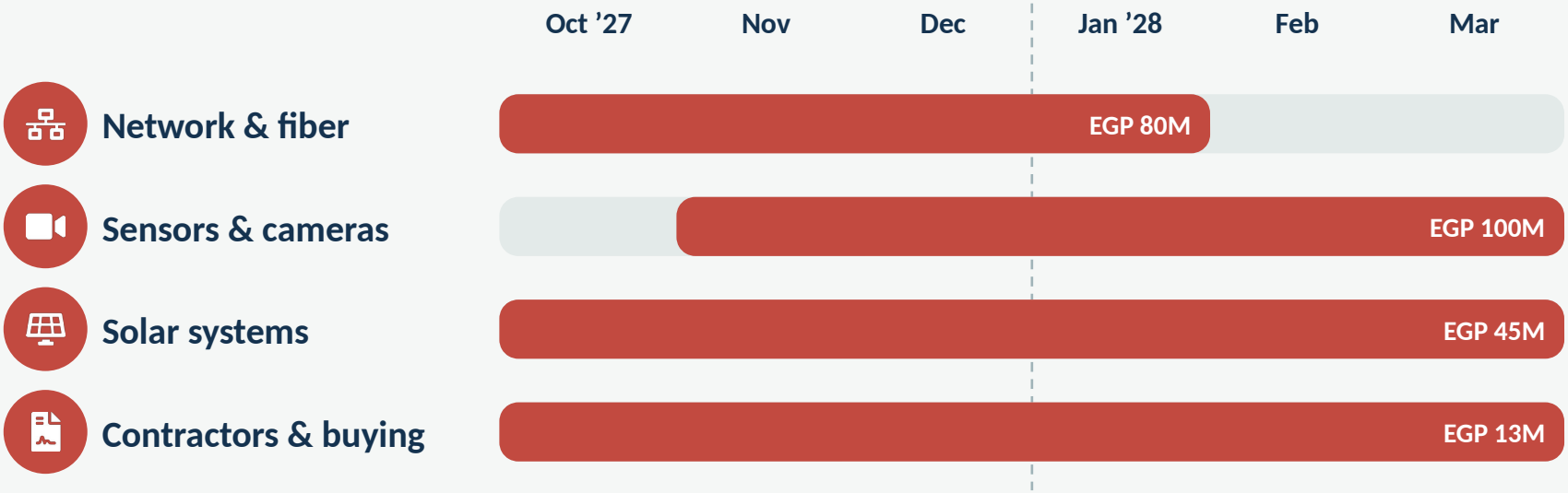
Dec 2028
Project closure

Each checkpoint needs the sponsor's approval before the next phase.

PHASE 4 · INFRASTRUCTURE

Planning Manager
Mariam Mahmoud Dabeesh

Inside the biggest phase



6 months

238M EGP · 43% of budget

186 people at peak

SMART TECHNOLOGY

Technical Manager
Rahma Khaled Ahmed

What we will build



IoT sensors

Measure traffic, energy & water



Smart cameras

Keep our streets safe



Smart traffic lights

Turn green when the road is empty



Solar energy

Clean energy, lower bills



Control center

The brain of the city



City app

City services on your phone

Technical Manager
Rahma Khaled Ahmed

How it all connects



1 · Collect

Sensors and cameras read the streets



2 · Send

The fiber network carries the data



3 · Decide

The control center acts, like turning a light green



4 · Show

Citizens see the result in the city app



Example: the road is empty → the light turns green → you get home sooner.

RESOURCE ALLOCATION

Technical Manager
Rahma Khaled Ahmed

Our five resources



People

186 at peak
(incl. 6 managers)



Materials

Fiber, cables,
manuals



Equipment

Sensors, cameras,
servers



Money

EGP 550M



Time

24 months

QUICK GAME

Technical Manager
Rahma Khaled Ahmed

Which technology fixes it?

- Stuck in traffic for an hour → ?
- A huge electricity bill → ?
- A long line to renew your ID → ?
- Help arrives too late after an accident → ?

? Tell us: which technology fixes each problem?

QUICK GAME · ANSWERS

Technical Manager
Rahma Khaled Ahmed

Which technology fixes it?

Stuck in traffic for an hour → Smart traffic lights

A huge electricity bill → Solar panels

A long line to renew your ID → The city app

Help arrives too late after an accident → Smart cameras + control center

Well done! Every problem has a smart solution.

BUDGET

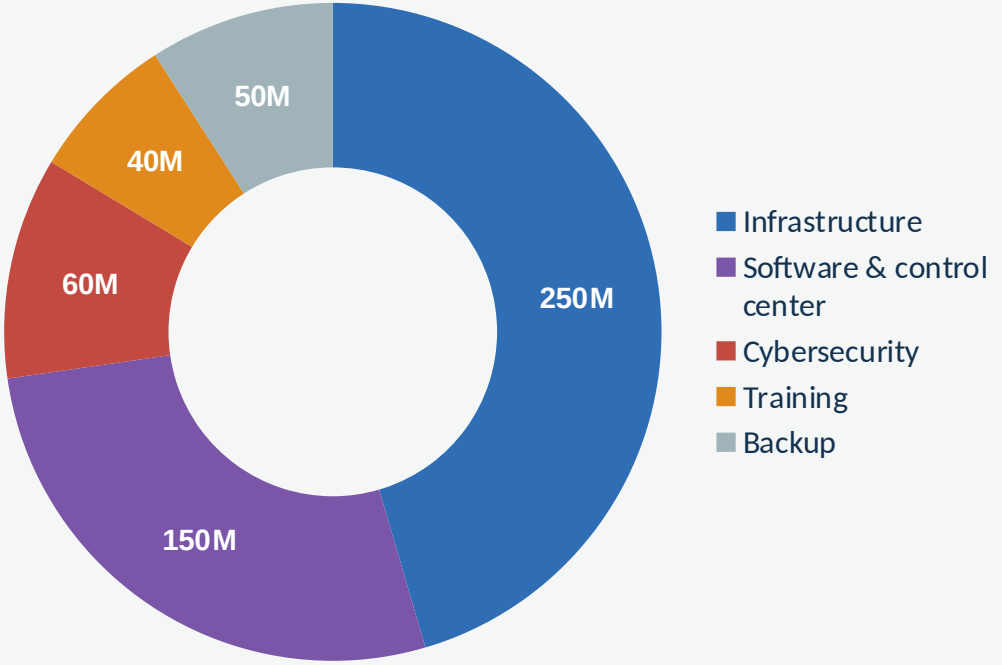
Where the money goes

Finance Manager
Moaz Hamed Ibrahim

550M

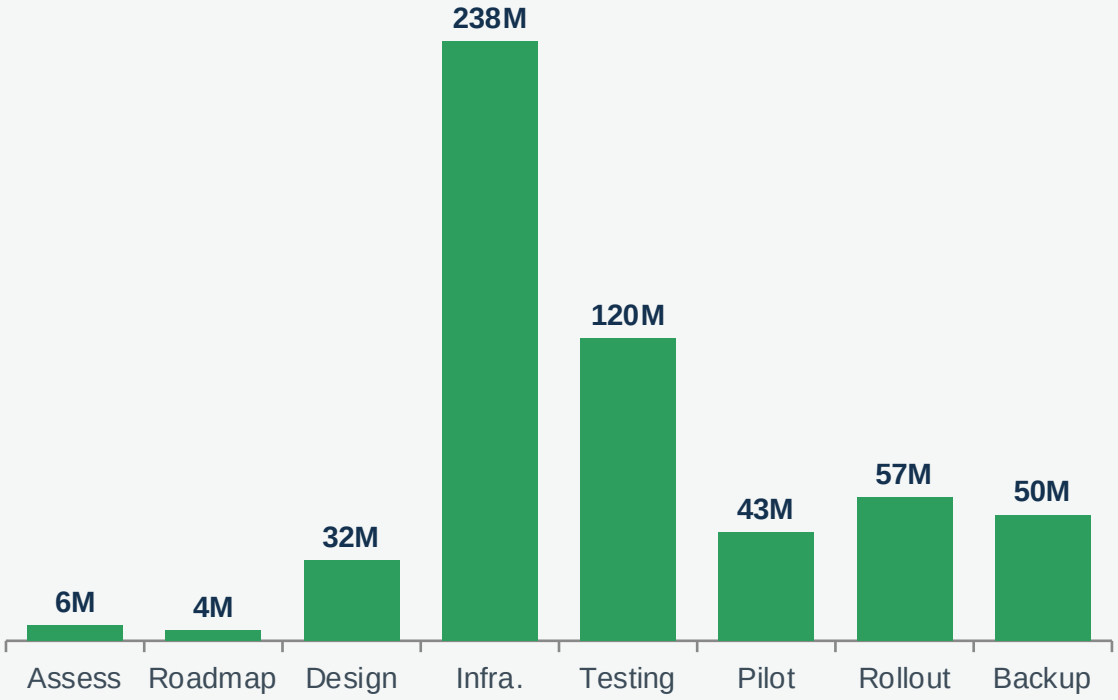
EGP total budget

including a 50M backup



BUDGET BY PHASE

When we spend it



Finance Manager
Moaz Hamed Ibrahim

43%

goes to Phase 4, infrastructure

Every month: budget vs. actual

More than 10% over plan? We flag it.

Finance Manager
Moaz Hamed Ibrahim

Where the money comes from

FUNDING

- Government budget
- Investors & partners
- 50M backup

REVENUE

- Land & offices
- Transport tickets
- Selling solar energy
- Data center space
- Smart services

RISK MANAGEMENT

Risk & Quality Manager
Omnia Sayed Abdelhakim

Our top risks

15

Cyberattack

Firewalls + 24/7 monitoring

12

Device failure

Test first, keep spares

12

Budget overrun

Monthly review + backup

12

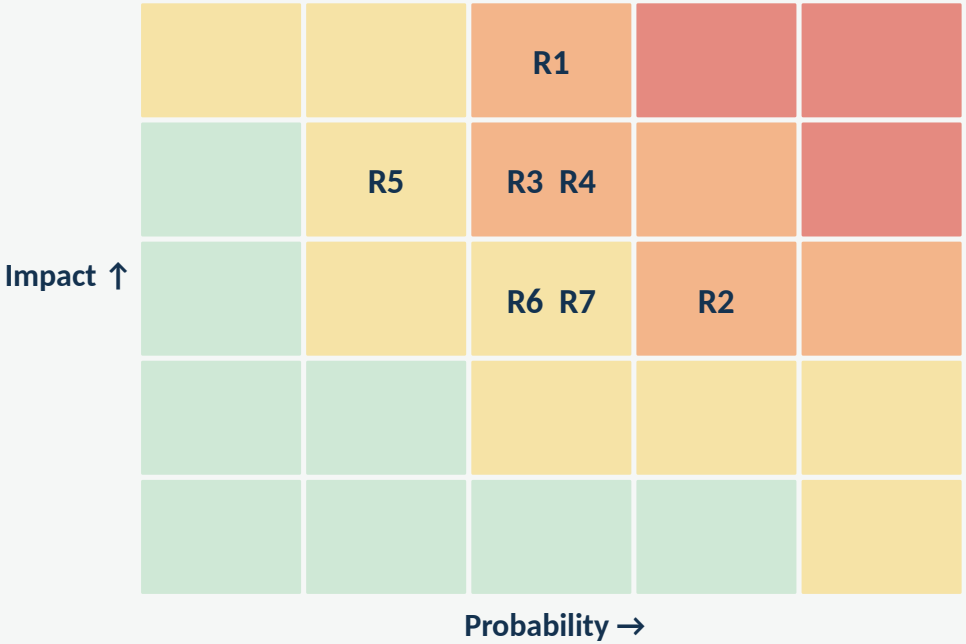
Data privacy

Clear privacy policy

Score = probability × impact. High risks are reviewed every week.

RISK MATRIX

Where our risks sit



R1	Cyberattack	15
R2	Device failure	12
R3	Budget overrun	12
R4	Data privacy	12
R6	Supplier delays	9
R7	Low app use	9
R5	Slow approvals	8

Risk & Quality Manager
Omnia Sayed Abdelhakim

QUALITY ASSURANCE

Risk & Quality Manager
Omnia Sayed Abdelhakim

How we protect quality

99%
system uptime

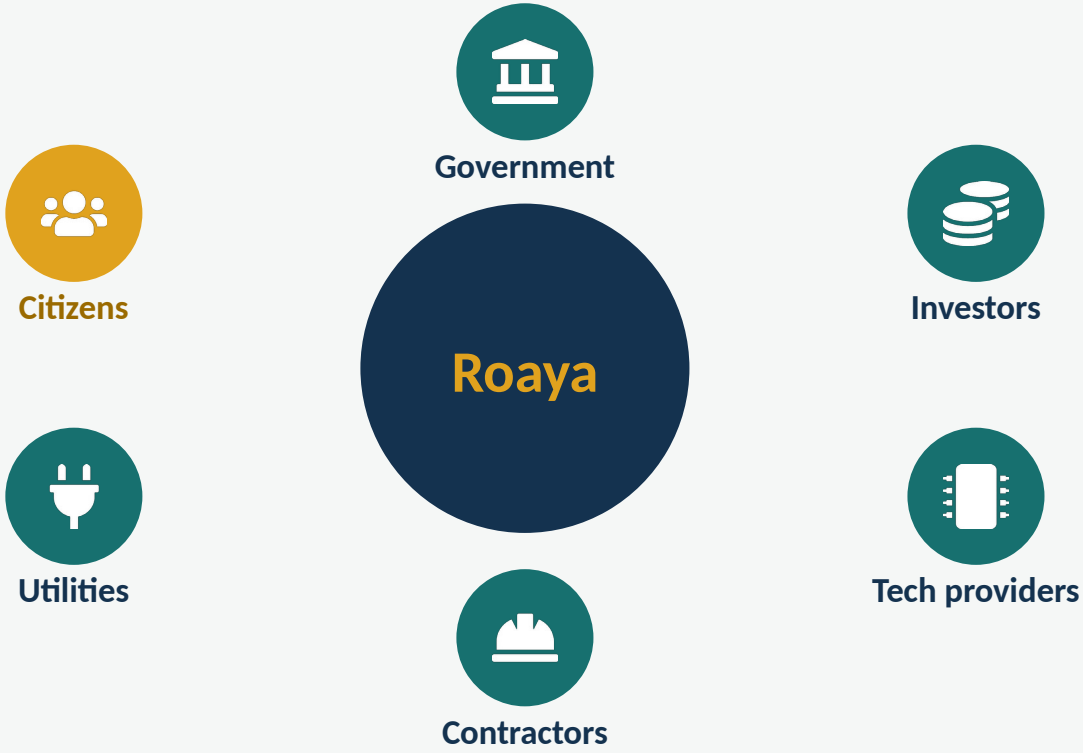
85%
citizen satisfaction

Checklist → Test → Fix issues → Final test → Approve

STAKEHOLDERS

Stakeholder Manager
Karim Mohamed Gad

Everyone we work with



STAKEHOLDER MAP

Power and interest

Stakeholder Manager
Karim Mohamed Gad

	Low interest	High interest
High power	KEEP SATISFIED Utilities	MANAGE CLOSELY Government Investors Citizens
Low power	MONITOR	KEEP INFORMED Contractors Tech providers

COMMUNICATION PLAN

How we stay in touch

Stakeholder Manager
Karim Mohamed Gad



Weekly

Team meeting



Every 2 weeks

Contractors



Monthly

Sponsor & investors



Monthly

Citizens: app & social media



24 hours

Critical issues to sponsor

Stakeholder Manager
Karim Mohamed Gad

What we learned



Start approvals early

In phases 1 and 2



Security from day one

Starting at the design phase



Pilot first

One district, then the whole city



Keep citizens involved

So they really use the services

QUICK GAME

True or false?

Whole team
All six managers

1. Roaya takes 24 months.
2. Phase 4 is the shortest phase.
3. Our traffic lights change by a fixed timer.
4. We keep 50 million pounds as a backup.
5. Our biggest risk is a cyberattack.
6. Egypt's New Capital has a smart control center.

Raise your hand if it's TRUE!

ROAYA رؤية

Thank you

Smart, green and connected.

Scan for all our files

Ministry of Communications and Information Technology

NTI
National Telecommunication Institute

